

Recording

Date	:	23/07/2026
Problem Specification	:	Find the Kth largest element from a list of integers entered by the user.
Assumption	:	Assume K is greater than 0.
Limitation	:	Duplicate values are considered when finding the Kth largest element.
Input	:	Number of elements (N) N integer values Value of K
Processing	:	Read the number of elements. Store the values in an ArrayList.. Sort the ArrayList in descending order. If valid, display the Kth largest element.
Output	:	The Kth largest element OR -1 if K is invalid..
Algorithm	:	Step 1: Read the number of elements N. Step 2: Create an empty ArrayList. Step 3: Read N integer values and store them in the ArrayList. Step 4: Read the value of K. Step 5: Sort the ArrayList in descending order. Step 6: Check whether the ArrayList is empty, $K \leq 0$, or K is greater than the size of the ArrayList. Step 7: If any condition is true, return -1. Step 8: Otherwise, return the element at index (K-1). Step 9: Display the result. Step 10: End the program
Programme listing	:	Programme file attached
Test data and expected output	:	1. Test data: N = 5 20 50 40 30 K=2 Expected output: 40

2. Test data: N = 4
 5 2 8 1
 K = 1
 Expected output: 8

3. Test data: N = 3
 10 20 30
 K = 5
 Expected output: -1

Output obtained for test data : Test Case 1 40
 Test Case 2 8
 Test Case 3 -1

Analysis : The numbers of operation required in performing the algorithm.

	+, -	/, *	%	</>/<=>=
Rad Input	N		-	-
Sort array list		-O(N log N)	-	

Conclusion : This program successfully finds the Kth largest element from a list of integers by sorting the ArrayList in descending order. It validates the input value of K before accessing the required element. If K is invalid or the list is empty, the program returns -1.

Discussion : The dominant operation in this program is sorting the ArrayList.